Yi Yang

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Professional Summary

Computer Science Ph.D. researcher developing graph representation learning and generative AI methods for molecular discovery, with publications in fragment-aware molecular learning and interpretable structure–property prediction. Experienced in 3D molecular generation, scientific LLM evaluation, and pharmaceutical R&D spanning cancer immunology and mRNA therapeutics.

Education

University of Texas at Dallas — Dallas, TX
Ph.D. in Computer Science, GPA: 3.8 — *Expected Dec. 2028*
– Erik Jonsson Graduate Scholarship in Computer Science, 2023

Ocean University of China — Qingdao, China
M.S. in Pharmaceutical Engineering — *2018*

Technical Skills

Machine Learning: Python, PyTorch, PyTorch Geometric, Transformers, GNNs, XGBoost, SHAP, pandas, NumPy, SQL
Scientific AI: RDKit, molecular representation learning, 3D molecular generation, ADMET prediction, structure–property modeling
LLM Systems & Evaluation: RAG, graph-structured memory, knowledge graphs, dense/sparse retrieval, structured outputs, LLM-as-a-judge, vLLM, Pydantic
Research Engineering: Slurm/HPC, Git, Docker/MLflow/FastAPI

Research Experience

Graduate Research Assistant | University of Texas at Dallas — Aug. 2024 – Present

PoseWords: SE(3)-Aware Generative Modeling for 3D Molecules — *May 2026 – Present*
– Designed a factorized 3D molecular language encoding fragment identity, conformation, and relative SE(3) pose.
– Building a graph-BPE pipeline for local-frame construction, pose encoding, and constrained autoregressive generation on GEOM-DRUGS and CrossDock.

BiScale-GTR: Fragment-Aware Graph Transformer [Paper] — *Oct. 2024 – May 2026*
– Developed a self-supervised Graph Transformer that contextualizes reusable graph-BPE molecular fragments with atom-level GNN embeddings.
– Designed chemically valid WL-hash tokenization with recursive out of vocabulary recovery; improved Peptides-func AP from 0.632 to 0.672 and Peptides-struct MAE from 0.293 to 0.262 using 78.5% less pretraining data than 2M-molecule baselines.

Interpretable ML for Organic Photovoltaic Property Prediction [Paper] — *Sep. 2024 – Aug. 2025*
– Built an RDKit-XGBoost pipeline for photovoltaic performance prediction across 350 donor molecules and six acceptors, achieving test $R^2 = 0.918$.
– Selected 10 predictors from 2,000+ descriptors and used SHAP to identify interpretable structure–property relationships; validated transfer to external datasets.

Selected Publications

- **Yi Yang**, Ovidiu Daescu. "BiScale-GTR: Fragment-Aware Graph Transformers for Multi-Scale Molecular Representation Learning." *Transactions on Machine Learning Research*, 2026.
- **Yi Yang**, Arowa Yasmeen, Ovidiu Daescu. "Uncovering Structure–Performance Relationships in Organic Photovoltaics: Interpretable ML Model for Predicting the Power Conversion Efficiency." *IEEE Journal of Photovoltaics*, 2025.

Selected Projects

HELP-E: Memory-Augmented LLM Agent for Multi-Session Reasoning [GitHub] — Mar. 2026 – May 2026
Llama-3.3-70B agent with graph-structured conversational memory — *Dallas, TX*
– Built and Dockerized a Llama-3.3-70B agent with FastAPI and vLLM, using typed graph memory and retrieval-selected context for multi-session concern tracking.
– Improved primary-concern F1 from 53.3% to 73.3% by encoding user beliefs, behavioral-change stages, and typed concern relations in a structured memory graph.
– Evaluated four system variants over 4,800 simulated dialogue turns using 30 multi-session user profiles, LLM-as-a-judge scoring, and trace-based error analysis.

Team Lead, BioTech Club at UTD | SQL Injection Detection with DistilBERT — Jul. 2024 – Jul. 2025
NLP security classification and production ML lifecycle — *Dallas, TX*
– Led an end-to-end pipeline over 100,000 curated SQL queries with reproducible preprocessing and data splits; fine-tuned DistilBERT to 0.91 F1, a 10% gain over classical ML baselines.
– Established an MLflow lifecycle with experiment tracking, registry-based promotion, reproducible packaging, and rollback; delivered CPU inference at 28 ms p50 / 46 ms p95 (batch size 1).

Industry Experience

Team Lead, StemiRNA Therapeutics — Apr. 2021 – Nov. 2022
Cancer Immunology – mRNA Vaccine Development — *Shanghai, China*
– Led a three-person wet-lab team supporting an mRNA cancer vaccine program that advanced to Phase II clinical evaluation.
– Gained working familiarity with preclinical development, including efficacy, toxicology, biodistribution, dose selection, and IND-enabling studies.